1.4 Limits and Continuity of Trigonometric, Exponential, and Logarithmic Functions

Name _____ Period ____ Date _____

AP Calculus 1.4 · Student sheet

I can…

- use the Squeeze Theorem to find a limit.
- use the limits of $\sin\theta/\theta$ and $(1-\cos\theta)/\theta$ to find trig limits.
- find limits of exponential and logarithmic functions using continuity.

Vocabulary · fill in the missing word

Squeeze Theorem — if $g(x) \le f(x) \le h(x)$ near c and g and h have the same _____ L at c, then the limit of f(x) at c is L

Two special limits — as $\theta \to 0$, $\sin\theta/\theta \to 1$ and $(1-\cos\theta)/\theta \to 0$ (θ in _____)

Part 1 · Radians matter

$\sin x / x$

x	±0.5	±0.1	±0.01
Radians			
Degrees			

Part 2 · Notes and examples (fill in each note, then work the example)

Squeeze Theorem: if $g(x) \le f(x) \le h(x)$ near c and g and h both approach L, then f(x) approaches _____.

Example 1

$-(x-2)^2 - 3 \le g(x) \le (x-2)^2 - 3$ for all x near 2. Find the limit of g(x) as $x \to 2$.

Special limits: as $\theta \to 0$, $\sin\theta/\theta \to 1$ and $(1-\cos\theta)/\theta \to$ _____.

Example 2

Find the limit as $\theta \to 0$ of $\dfrac{\sin(5\theta)}{2\theta}$.

Example 3

Find each limit as $x \to 0$.

a. $\dfrac{\sin(3x)}{\sin(2x)}$

b. $\tan x / x$

Example 4

Find the limit as $x \to 0$ of $x^2 \sin(\frac{1}{x})$.

Continuity: sin x, cos x, and e^x are continuous everywhere; tan x and ln x are continuous on their _____.

Example 5

Find each limit.

a. as $x \to \pi$: $\sin x + \cos x$

b. as $x \to 1$: $\ln(x^2 + e - 1)$

Part 3 · Partner practice

Partner 1

Find the limit as $x \to 0$ of $\frac{\tan(4x)}{3x}$.

Partner 2

(AP® Practice) The limit as $x \to 0$ of $\frac{\sin(4x)}{2x}$ is

(A) 0 (B) $\frac{1}{2}$ (C) 2 (D) 4